Software Design

What is Software Design?

Requirements vs. Design

Requirements	Design
Describes **what** the system should do	Describes **how** the system will do it
Focuses on user/business needs	Focuses on technical solution
Usually technology-independent	Often technology-specific
Defines expected behavior	Defines system structure and implementation approach
Example: Students shall search books	Example: Use a search service connected to the book database

Example

- **Requirement:**

  The system shall allow students to search books by title, author, ISBN, and category

- **Design decision:**

  Implement a *BookSearchService* that receives search criteria, queries the *Books* database table, and returns matching records to the user interface

Requirement: "What should happen?"

Design: "How will we make it happen?"

Major Activities in the Design Phase

i. SRS

ii. Identify Major Components

iii. Architectural Design

iv. Database/Data Design

v. Interface Design

vi. Component/Detailed Design

vii. Design Review

Architectural Design

- Architectural design defines the **high-level structure of the software system**.
- It identifies:
  - Major components
  - Subsystems
  - Layers
  - Relationships between components
  - Communication mechanisms